1. Complex research agents depend on retrieval

RAG fetches the enterprise domain knowledge an agent needs to reason

- Foundation models do not contain every organization’s **current, private, or regulated evidence**.
- RAG is the agent’s evidence supply chain: it selects what enters the reasoning context.
- Poor retrieval creates confident conclusions from missing, stale, or incorrectly grouped evidence.
- In drug research, **product, strength, route, dosage form, and regulatory status** all matter.

> **If retrieval is wrong, better generation cannot rescue the answer.**

Goal: make enterprise retrieval observable, testable, and self-improving.

2. FDA Knowledge Graph

Unifying fragmented regulatory evidence for precise retrieval

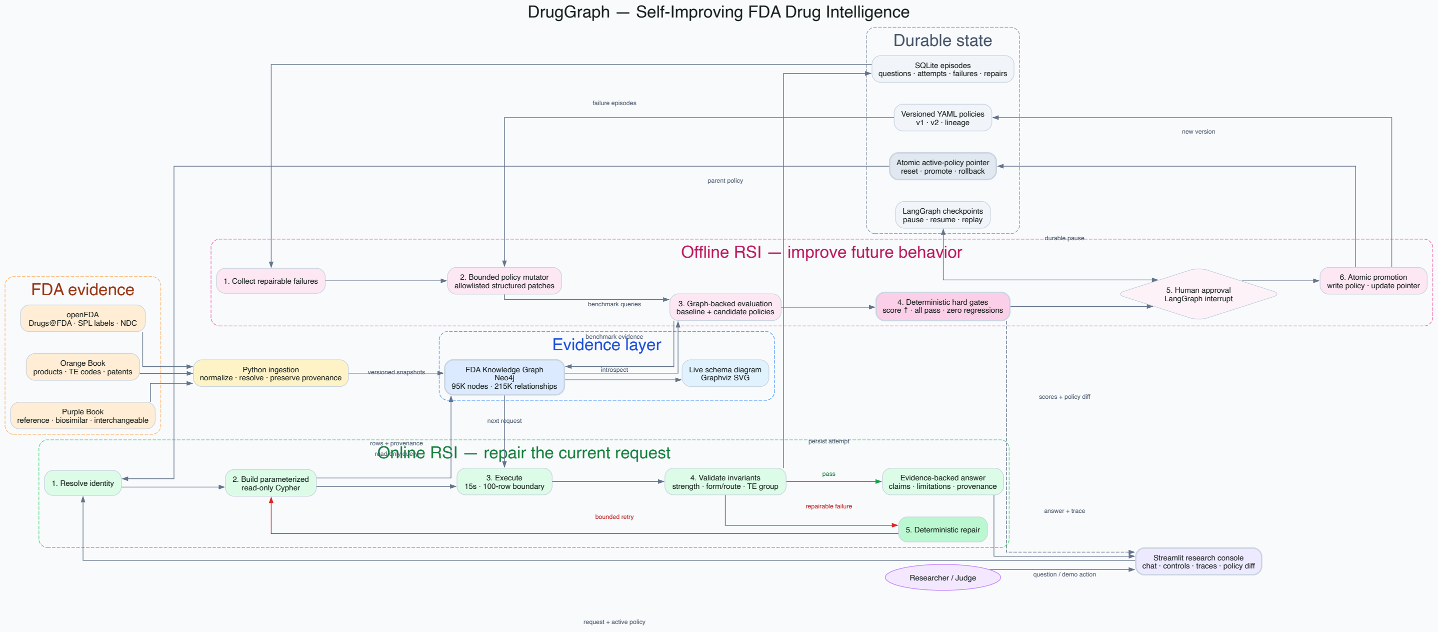
Source	Evidence
Orange Book	Products, applications, RLD/RS status, TE codes, patents
Purple Book	Reference biologics, biosimilars, interchangeable products
openFDA	Drugs@FDA, SPL labels, and NDC listings

The graph connects

Drug names → products → applications → regulatory evidence → labels

- Approximately **95K nodes** and **215K relationships**
- Provenance remains attached to every claim
- State substitution law remains a separate, versioned source

3. DrugGraph system design



- **Neo4j:** FDA evidence and provenance
- **LangGraph:** online repair and offline optimization
- **SQLite + YAML:** durable episodes, checkpoints, policy lineage, and rollback

4. Self-improving retrieval—not self-modifying code

```
Detect failure → Repair request → Record episode
       ↓
Generate bounded policies → Run graph–backed evals
       ↓
Hard safety gates → Human approval → Promote → Replay
```

Demonstrated policy evolution

	Policy v1	Promoted v2
Evaluation score	**37.5%**	**100%**
Lipitor query attempts	**2**	**1**
Adalimumab capability	Blocked	Enabled after Purple Book gate

Promotion rule: higher score + all hard checks + zero regressions + approval. Every mutation is allowlisted and shown as a YAML diff.

5. What worked—and what comes next

What worked

- Deterministic FDA invariants caught errors that an LLM judge might miss.
- Graph-backed evaluation made policy improvement measurable.
- Approval, versioning, reset, and rollback made the loop demoable and auditable.

Next

- Add an **LLM judge** for usefulness and clarity—never as the sole promotion gate.
- Add an LLM proposer that emits only structured, allowlisted policy patches.
- Evaluate on frozen FDA graph snapshots and shadow production traffic.
- Expand RSI to label monitoring, biologic navigation, and sourced state-law rules.

> **DrugGraph turns retrieval failures into tested, durable improvements.**